CSE 107 — Midterm 1 Practice

Solutions Fall 2026

These solutions show the level of detail expected. Equivalent correct methods are acceptable, and numerical expressions may be left unevaluated.

1.

(a) **False.** On five equally likely outcomes, let $A = \{1, 2\}$ and $B = \{1, 2, 3\}$. Then $P(A) = 2/5$ and $P(B) = 3/5$, but the events overlap.

(b) **True.**

(c) **True.**

(d) **False.** The probabilities must also sum to 1.

(e) **True.**

(f) **False.** The outcomes need not be equally likely, so counting them alone does not determine $P(A)$.

(g) **True.**

(h) **False.** In a binomial model, the success probability is the same on every trial.

2.

(a) The possible values are $-1, 1, 3, 4$. Each probability mass is the increase in the CDF at that value:

x	-1	1	3	4
$p_X(x)$	$1/10$	$3/10$	$2/5$	$1/5$

$p_X(x) = 0$ otherwise.

(b)

$$P(X < 1) = P(X = -1) = \frac{1}{10},$$
$$P(-1 < X \leq 3) = F_X(3) - F_X(-1) = \frac{4}{5} - \frac{1}{10} = \frac{7}{10}.$$

3.

(a) One letter appears twice and the other appears once. Choose the repeated letter, the other letter, and the position of the letter that appears once:

$$5 \cdot 4 \cdot 3 = 60.$$

(b) Given that the endpoints differ, there are $5 \cdot 4 \cdot 5$ equally likely codes. To use exactly two different letters, the middle letter must match one of the two endpoints. Hence the conditional probability is

$$\frac{5 \cdot 4 \cdot 2}{5 \cdot 4 \cdot 5} = \frac{2}{5}.$$

4.

(a)

$$P(E \cap D) = P(D \mid E)P(E) = \frac{3}{5} \cdot \frac{2}{5} = \frac{6}{25},$$
$$P(D) = \frac{P(E \cap D)}{P(E \mid D)} = \frac{6/25}{4/5} = \frac{3}{10}.$$

(b)

$$P(D \mid E^c) = \frac{P(D) - P(D \cap E)}{1 - P(E)} = \frac{3/10 - 6/25}{3/5} = \frac{1}{10}.$$

(c) The two disjoint cases are $E \cap D^c$ and $E^c \cap D$. Their total probability is

$$P(E) + P(D) - 2P(E \cap D) = \frac{2}{5} + \frac{3}{10} - 2\left(\frac{6}{25}\right) = \frac{11}{50}.$$

(d) The events are not independent, since $P(E \mid D) = 4/5 \neq 2/5 = P(E)$.

5.

(a) There are six independent trials with the same error probability, so

$$N \sim \text{Bin}\left(6, \frac{1}{5}\right), \qquad N \in \{0, 1, 2, 3, 4, 5, 6\}.$$

$$\begin{aligned} P(N \geq 5) &= P(N = 5) + P(N = 6) \\ &= \binom{6}{5}\left(\frac{1}{5}\right)^5\left(\frac{4}{5}\right) + \left(\frac{1}{5}\right)^6 = \frac{1}{625}. \end{aligned}$$

(b) Every set of two incorrect readings has probability $(1/5)^2(4/5)^4$, so the $\binom{6}{2}$ possible pairs are equally likely given $N = 2$. Both incorrect readings must be among the last four:

$$P(\text{first two correct} \mid N = 2) = \frac{\binom{4}{2}}{\binom{6}{2}} = \frac{2}{5}.$$

(c) Independence gives

$$P(\text{all six correct}) = \left(\frac{4}{5}\right)^5\left(\frac{1}{2}\right) = \frac{512}{3125}.$$

The binomial model is not valid here because the error probability is not the same on every reading.